

Computer Graphics

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This generally refers to creation, manipulation and storage of models and images on a computer system. A graphic is an image or visual representation of an object. Therefore, computer graphics is simply an image created, stored and displayed on a computer screen. The end product of the computer graphics is a picture it may be a business graph, drawing, or engineering.

Computer graphics are used for the creation, development of videos and computer programs, science modelling, catalog design and other commercial art.

Graphic Image - is a photo, painting, sketch, or other image that provides a detailed representation of an object, scenario, person, animal, chart or graph among other things.

Various Representations of Graphics

- **Picture:** abbreviated as pic, a picture is a visual capture of an object. Picture can be created using devices such as digital camera, scanner, smartphone, or are works of art created on the computer.
- **Drawing:** refers to the work you create in an app such as CorelDraw: for example, custom artwork, logos, poster, and newsletters.
- **Chart:** A chart is a graphical representation for data visualization, in which "the data is represented by symbols, such as bars in a bar chart, lines in a line chart, or slices in a pie chart". A chart can represent tabular numeric data, function or some kinds of quality structure and provides different information.
- **Animation:** Animation refers to a graphic element that displays movement. When you apply some visual effects to the content in slide or to the entire slide then the outcome is known as animated slide. A good example of application package that deal with animation is known as (Microsoft PowerPoint).

Computer Graphic Applications

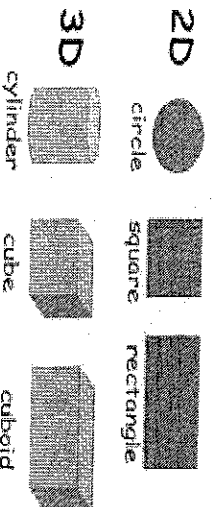
These are applications or softwares used to create a computer graphic. Examples of CG applications include;

- Adobe Photoshop
- Adobe Illustrator
- AutoCAD
- Corel Draw
- Microsoft Desktop publisher
- Paint Shop Pro

Classification of Computer graphics

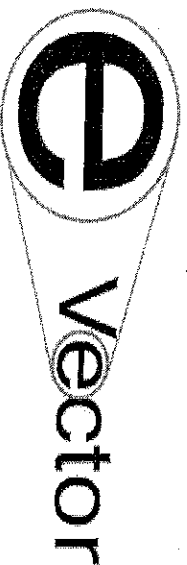
Computer Graphics is basically classified into two (2) types.

1. 2D Graphics
2. 3D Graphics



1. 2D Graphics: Meaning “Two Dimensional” graphics is often used in applications that were first developed around traditional printing and drawing technologies. Typography, cartography, technical drawing, and advertising. only tools like lines, shapes it's joining to create object. 2D graphics appear flat on the screen and are viewed from only one angle. They are of two (2) types;

- **Vector Graphics:** These type of Graphics mainly uses “Lines and Shapes” and text to make images or even complex images. Vector graphics are made up of paths, which may be lines, shapes, letters, or other scalable objects. They are often used for creating logos, signs, and other types of drawings. Unlike raster graphics, vector graphics can be scaled to a larger size without losing quality. Vector Graphics applications include programs such as Inkscape, Adobe Illustrator, Corel Draw, AutoDesk Sketchbook, Affinity Designer, and Adobe fresco.

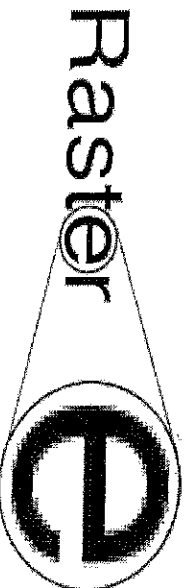


Features of a vector graphic

- It can be re-sized infinitely (larger or smaller) and will still print out just as clearly, with no increase (or decrease) in file size.
- Vector files are thin in size since they have less information than bitmap image.

Common Vector File Types: AI, EPS, SVG, WMF, EMF

- **Raster Graphics:** Raster graphics, also called bitmap graphics, they generally use “Pixels” to make images. It is made up of a collection of tiny, uniformly sized pixels, which are arranged in a two-dimensional grid made up of columns and rows. Each pixel contains one or more bits of information, depending on the degree of detail in the image. Raster Graphics often use some programs like Corel Paint Shop Pro and Adobe Photoshop, Google picasa, GIMP, etc



A pixel is a single point or the smallest single element in a raster image display device. If you zoom in to a raster image you may start to see a lot of little tiny squares these tiny squares are the pixels. *Common Raster Image Types: JPG, TIFF, GIF, PNG, and BMP.*

Example of raster graphics include pictures from digital camera, optical scanned images and digital images on the internet.

Features of Raster Image

- They are resolution dependent.
- The number of pixels in an image is per inch, & it determine the quality of an image.
- The higher the resolution of an image the higher quality so also lower viz a viz.
- If you enlarge it without changing resolution, it will lose quality and look blurry.

2. 3D Graphics

3D Graphics "Three Dimensional" graphics has parameters like Height, Length and Depth. As this consists of various parameters, ultimate resulting image looks like *real objects*. It is sometimes called 3DCGI. It operates on three coordinates X and Y and Z coordinates, unlike 2D which operates on two (2) coordinates X and Y.

Applications areas of Computer Graphics

1. **Computer Aided Design(CAD)** - Computer graphics are mainly used in the design process, these applications are especially used in the Architectural, engineering field. For example, making buildings and other structural designs, designing automobiles and aircraft, ships, etc.

Through CAD software, you can prepare the surface in 2D and 3D form, so that you can see any part of the surface from any angle. CAD (Computer-aided design) and CADD (Computer-aided drafting and design) are mainly used for designing automobiles, aircraft, spacecraft, Shippers, computers, and home appliances.

2. **Computer Aided Learning (CAL)**: Today, a computer graphics tool is used to create large-scale Teaching Seminars and Conference Presentations. Microsoft *PowerPoint* and *Google slides* are being used more today for creating presentations. Here "Presentation" refers to the act of presenting important points of a topic to an audience for a lecture or a Potential customer for a new product. It can also be used for

- Preparation of reports eg. financial, statistical, mathematical, scientific, economic data for research & managerial reports,
- Creation of bar graphs, pie charts, time chart, can be done using the graphic tools present in computer graphics.

3. **In Education** - In today's era, graphics are also being used in the education field. Today teachers are also teaching their students through graphics. By teaching children through graphics, children are able to understand that topic very well because it has been found in a study that by reading through graphics, children understand quickly and remember what they have read for a long time. By teaching the child with graphics, children read by understanding cartoons, and their interest in studies increases.

4. **Entertainment**: Graphics are also being used in the field of entertainment, due to which people are getting more attracted towards entertainment. Graphics are used to create motion films, television shows and cartoons. The use of graphics in movies makes the audience feel like a real fantasy of watching a movie.

- Watching 3D films and Animation movies.
- Children also like to watch cartoons.
- Use of graphics apps on smart gadgets to makes videos and photos more attractive.

5. **Computer Games**: In the game industry where focus and interactivity are the key players, computer graphics helps in providing such features in the efficient way. These games require human interaction with the User Interface for making nice visual result on the Video Device. A video game is an electronic game that involves interaction with a user interface to generate visual feedback on a raster display device. The electronic systems used to play video games are known as platforms. This platform is created through graphics.

6. **Training** - In the field of training graphics are used so that the trainee can get the real experience of learning. **For example** –
 - Simulator is a device which is used for training either aeroplane, car, train, Rocket etc. The simulator gives you real driving experience through graphics. You will feel as if you are driving a real car/aeroplane etc, There are many types of simulators like - flight simulator very much needed for training pilots to understand easily, four wheeler simulator etc.
 - It is simple and very useful to build training modules with computer graphics.
7. **Industrial Design (ID)**: Industrial graphic designers use graphics apps to help companies cope with product design on an industrial scale. the professional practice of designing products, devices, objects, and services used by millions of people around the world every day. Graphic apps like Catia, LabView, AutoDesk Inventor, (PTC CreoSuite), ANSYS, SolidWorks, Rhino 7, RhinoCAM for Rhino, Adobe Creative Suite, Keyshot, Adobe Sketchbook Pro (free), Autodesk Fusion 360, Arduino IDE (free), Autodesk Meshmixer (free), Meshroom (free).
8. **Image Processing (photography)** - Specific types of photos or photographs need to be edited to be used at various locations for various reasons. One of the many techniques of computer graphics is to transform existing images into optimized ones to enhance their understanding.
9. **Image Processing (Medical)**: - In Medical field, concerning Image Processing CG is used in various technologies to interpret already existing pictures and is useful to modify Photographs and TV scans. Basically CG is used to improve the picture quality, and visualizing effects. Some of its applications include
 - i). Tomography
 - ii). Ultrasonic medical scanners
 - iii). Disease Picture enhancements
10. **Graphics User Interface GUI**: Using computer graphics, we can design the user interface of any software, web application, website, mobile app. Whatever website or web application you to see or when you visit any website or web application and the interface you see is all made using graphics. Or you use any mobile you use, its interface is also made using graphics. CG is used to make Menus, Icons (Graphical Symbols), to make window manager. Graphics play an important role in creating the interface of any software.
11. **Computer Art**: Using computer graphics, you can also create commercial art so that the product looks more attractive in appearance. So that that company can promote the graphics image of its product like Graphical poster, brand logo design for companies, institutions and events, cartoon poster, product advertisement. Using graphics designing, you can create any type of graphics image like graphics template, YouTube thumbnail, post feature image, Facebook post image, birthday card and many more.
12. **Cartography**: Cartography is a discipline that deals with the generation of informative maps and charts. Computer graphics are widely used to craft and design the map for different purposes. Computer graphics help to design the map focused on different strategies. As cartography combine science, aesthetics, and technique, it crafts maps so that user can easily communicate with map and gather necessary information.

Digital Image File Format

The list below outlines some of the more common file types and provides a brief description, how the file is best used, and any special attributes the file may have.

1. **JPEG (.jpg, .jpeg):** which stands for Joint Photographic Experts Groups is a "lossy" format meaning that the image is compressed to make a smaller file. The compression does create a loss in quality but this loss is generally not noticeable. JPEG files are very common on the Internet and JPEG is a popular format for digital cameras - making it ideal for web use and non-professional prints.
Category: Raster Image file
Compression: Lossy - some file information is compressed or lost
Best For: Web Images, Non-Professional Printing, E-Mail, Powerpoint
Special Attributes: Can choose amount of compression when saving in image editing programs like Adobe Photoshop or GIMP.
2. **GIF (.gif):** GIF or Graphics Interchange Format files are widely used for web graphics, because they are limited to only 256 colors, and can allow for transparency, and can be animated. GIF files are typically small in size and are very portable.
Category: Raster Image file
Compression: Lossless - compression without loss of quality.
Best For: Web Images.
Special Attributes: Can be Animated, Can Save Transparency.
3. **PNG (.png) :** PNG or Portable Network Graphics files are a lossless image format originally designed to improve upon and replace the gif format. PNG files are able to handle up to 16 million colors, unlike the 256 colors supported by GIF.
Category: Raster Image file
Compression: Lossless - compression without loss of quality.
Best For: Web Images.
Special Attributes: Save Transparency.
4. **EPS (.eps) :** An EPS or Encapsulated PostScript file is a common vector file type. EPS files can be opened in many illustration applications such as Adobe Illustrator or CorelDRAW.
Category: Vector Image file
Compression: None - uses vector information
Best For: Vector artwork, illustrations
Special Attributes: Saves vector information.
5. **RAW Image Files (.raw, .cr2, .nef, .orf, .sr2, and more)**
RAW images are images that are unprocessed that have been created by a camera or scanner. Many digital SLR cameras can shoot in RAW, whether it be a .raw, .cr2, or .nef. These RAW images are the equivalent of a digital negative, meaning that they hold a lot of image information, but still need to be processed in an editor such as Adobe Photoshop or Lightroom.
Category: Camera raw files

Compression: None

Best For: Photography

Special Attributes: Saves metadata, unprocessed, lots of information.

6. **TIFF (.tif, .tiff) :** TIFF or Tagged Image File Format are lossless images files meaning that they do not need to compress or lose any image quality or information (although there are options for compression), allowing for very high-quality images but also larger file sizes.

Category: Raster Image files

Compression: Lossless - no compression. Very high-quality images.

Best For: High quality prints, professional publications, archival copies

Special Attributes: Can save transparencies.

7. **Bitmap (.bmp) :** BMP or Bitmap Image File is a format developed by Microsoft for Windows. There is no compression or information loss with BMP files which allow images to have very high quality, but also very large file sizes. Due to BMP being a proprietary format, it is generally recommended to use TIFF files.

Category: Raster Image file

Compression:

Best For: High quality scans, archival copies

None

8. **SVG (.svg):** Scalable Vector Graphics (SVG) is a web-friendly vector file format. As opposed to pixel-based raster files like JPEGs, vector files store images via mathematical formulas based on points and lines on a grid.

Category: Vector Image files

Compression: None - uses vector information

Best for: Website icon and logos, Animations